

Darren Zheng

 [Personal Website](#)

 darrenzheng2480@gmail.com

 [LinkedIn](#)

EDUCATION

University of Pennsylvania, College of Arts and Sciences

B.A. in Mathematics, Minor in Statistics & Data Science, Ben Franklin Scholar

May 2025

B.A. GPA: 3.51/4.0

MMI Preparatory School

High School Diploma

June 2021

PROJECTS

Independent Study

Sept 2025 – Dec 2025

- Reading "Fourier Analysis & Hausdorff Dimension" by Pertti Mattila under Prof. Gressman
- Short write-up regarding the proof of the Wolff-Erdogan Theorem can be found on my personal website [here](#)

Undergraduate Honor's Thesis | [Link to PDF](#)

June 2024 – May 2025

- Constructed intrinsic Morse functions on Grassmannian manifolds $\mathbf{G}_k(\mathbb{R}^n)$ to compute singular homology groups under Prof. DeTurck

Directed Reading Program | [Link to Project Write-ups](#)

Aug 2022 – May 2025

- Semester-long independent study with a graduate student. Prepared a presentation/write-up at the end of each semester to rest of participants and mentors. Participated over 6 semesters
 - * Knot Invariants using Algebraic Topology (Yi Wang)
 - * Understanding Generating Functions (Xinxuan Zhang)
 - * Pontryagin Duality of LCA Groups (Travis Leadbetter)
 - * The Distinct Distances Problem (Jae Ho Choi)
 - * Besicovitch Sets & Hausdorff Dimension (Hunter Stufflebeam)
 - * Studying Mapping Class Groups (Ellis Buckminster)

EWS Detection using Topological Data Analysis | [PDF Link](#)

Jan 2024 – Apr 2024

- Applied persistent homology techniques to time-series data to develop early warning signals for market crashes
- Implemented Python pipeline (`scikit-tda`, `risper`, `persim`) to transform historic stock index data into persistence landscapes and diagrams for sliding window analysis
- Conducted statistical validation using Mann-Kendall test to analyze significance of methodology

WORK

Private Competitive Mathematics Coaching | *Tutor*

May 2025 – Present

- Guide 14 students with individualized training for math contests, such as AIME, USAMO, and MathCounts
- Prepare personalized problem sets & mock exams to target technical gaps based on past student performance

University of Pennsylvania Math Department | *Teaching Assistant*

Aug 2022 – May 2025

- Developed and delivered weekly recitations for 5 proof-based calculus and linear algebra courses
- Classes: Math 114 (Calculus II) x2, MATH116/260 (Honors Calculus I/II), MATH312 (Linear Algebra)
- Graded homework/exams, providing feedback on proof-writing; Received 2 Departmental teaching awards

Ross Mathematics Program | *Administrator*

June 2024 – July 2024

- Managed operations for a 6-week residential summer math camp for ~80 high schoolers and ~20 instructional staff
- Presented weekly mathematical talks, involving formal logic, algebraic and point-set topology to advanced students

AwesomeMath Summer Program | *Teaching Assistant*

June – Aug 2022/2023/2025

- Instructed six 3-week-long courses of ~15-30 high school students each summer and taught primarily number theory, combinatorics, and abstract algebra, facilitating problem sessions and active learning

ACTIVITIES

Penn Math Contest Club | *Co-founder & President*

Aug 2021 – May 2025

- Founded student club that organizes an on-campus high-school mathematics contest with original problems
- Planned event logistics, coordinating with campus facilities & training ~30 club members/volunteers; Secured corporate sponsorships to offset operational expenses and created handbooks for future leadership
- Conducted marketing outreach, growing from ~40 online contestants (COVID) to ~170 in-person and ~200 online in Spring 2025

Penn Undergraduate Math Society | *President*

Jan 2024 – May 2025

- Led Penn's undergraduate math club, directing a board of ~10 to support math-adjacent programming
- Organized multiple events (~13 per semester), including major/course advising, research talks, integration bee, merch sale, faculty lunches, DRP x PUMS collaboration, and industry panels

Putnam Club | *Organizer*

Aug 2021 – Dec 2024

- Organized club that meets weekly on Sundays in the fall to practice for the Putnam Exam

TEACHING

University of Pennsylvania Math Department | *Teaching Assistant*

Aug 2022 – May 2025

- Responsible for grading assignments/exams, weekly office hours and recitations for most classes
 - * Grader for MATH3200 - Linear Algebra (Spring 2025)
 - * TA for MATH1410 - Calculus II (Fall 2024)
 - * TA for MATH2600 - Honors Calculus II (Spring 2024)
 - * TA for MATH1610 - Honors Calculus I (Fall 2023)
 - * TA for MATH1410 - Calculus II (Fall 2022)

VOLUNTEERING

Science Olympiad at Penn | *Test Writer*

Aug 2021 – May 2025

- Wrote tests for It's About Time, Fermi Problems, Optics, and Codebusters categories for high school competitors over 4 years
- Joined finance board during 2023 — 2024 to secure sponsorship funding for the high-school contest

Matriculate | *Mentor*

Aug 2022 – May 2024

- Mentored first-generation and low-income high school seniors through the year-long college admissions process
- Scheduled weekly meetings and oversaw 3 cohorts (12 students in total)

Princeton Prison Teaching Initiative | *Grader*

Jan 2024 – May 2024

- Graded HW + Exams for a statistics class

Bridge to Math | *Volunteer*

Aug 2022 – May 2025

- Hosted some additional quiz/finals prep sessions throughout the semester as organized by Bridge-to-Math club @ Penn

AWARDS

Good Teaching Award for MATH2600

May 2024

Good Teaching Award for MATH1610

Dec 2023